

Abel Bekana

Software Engineer

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PROFILE

Software Engineer with strong foundations in algorithms, data structures, and system design, combined with hands-on experience in backend development and machine learning systems. Experienced in building scalable applications and designing intelligent systems, with a strong interest in evaluating and improving AI generated code through rigorous problem solving and technical analysis.

PROFESSIONAL EXPERIENCE

Backend & ML Engineer <i>Vertex</i>	2023 – 2024 remote
Architected an adaptive coding-challenge platform using FastAPI, PostgreSQL, and a custom reinforcement-learning model that recommended questions in real time, reducing user drop-off by 40%.	
Machine Learning Engineer <i>Innowise</i>	2022 – 2023
- Designed scalable data pipelines for preprocessing and feature engineering on AWS S3 and Dockerized environments, reducing training time by 35%. - Integrated real-time inference APIs with backend systems, improving response latency by 40% and system reliability across 10K+ daily requests.	
Backend & ML Engineer <i>Revelo</i>	2025
- Evaluated AI-generated code and system outputs for correctness, efficiency, and production readiness, ensuring alignment with software engineering best practices. - Analyzed model responses across diverse programming scenarios, identifying logical errors, edge-case failures, and optimization opportunities. - Provided structured feedback and grading on AI outputs, improving overall model reliability and real-world applicability	

EDUCATION

Bachelors of science in Software Engineering <i>Addis Ababa Science and Technology University</i>	2020 – 2024
Relevant Course Work: Machine learning, Artificial intelligence, Data structure and Algorithm, Operating system, Human computer Interaction	

TECHNICAL STRENGTHS

- Algorithms & Data Structures (problem solving)
- System Design & Backend Architecture
- Code Debugging & Optimization
- API Design & Distributed Systems
- AI/ML Systems Evaluation & Model Behavior Analysis

PROJECTS

HealthInsight
- Developed a machine learning model using XGBoost and TensorFlow to predict patient readmission risks with 90% accuracy across 50K+ medical records. - Built automated ETL and preprocessing pipelines with pandas and Airflow, reducing data preparation time by 40%. - Deployed the model via FastAPI and Docker, serving real-time predictions to hospital dashboards with <300 ms latency.
VisionGuard
Designed and deployed an object detection and anomaly recognition system using YOLOv8 and OpenCV, achieving 95% precision in identifying suspicious activity across 120+ camera feeds.

- Integrated Kafka streaming for real-time event processing and alerts, maintaining <200 ms inference latency on edge devices.

Ethical Bias Mitigation in Large Language Models

- Conducted a study on algorithmic fairness and bias correction in transformer-based models. Implemented reweighting and counterfactual data augmentation techniques, reducing bias score by 35% across gender and age groups.